

Zachary Hoheb

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EDUCATION

University of Michigan

B.S. in Computer Science and Mathematics, LSA Honors Program

Ann Arbor, MI

Graduation: May 2028

- **GPA:** 4.0/4.0 | **Honors:** Branstrom Freshman Prize (top 5%), James B. Angell Scholar, University Honors
- **Coursework:** Data Structures & Algorithms, Computer Organization, Foundations of Computer Science, Web Systems, Machine Learning, Discrete Math, Probability, Linear Algebra
- **Planned:** Operating Systems, Distributed Systems, Database Management Systems, Software Engineering

EXPERIENCE

Altheros Capital – Midwest Health (Telehealth Platform)

December 2025 – August 2026

Full-Stack Software Engineering Intern

Remote

- Architected and shipped a Provider Availability Management system from scratch: a RESTful Node.js/Express API (bulk scheduling, recurring windows, out-of-office blocking), PostgreSQL schema, and React/TypeScript calendar UI, now used in production by 80 therapists and replacing a manual process that took 10+ hours/week
- Implemented authentication and authorization with AWS Cognito MFA and role-based access control across API routes and UI views, and integrated Stripe Elements for secure in-app payments in the patient billing flow
- Resolved a 2-day production outage blocking all patient logins, tracing it through Docker logs, Postman, and PostgreSQL to UUID validation errors, unregistered routes, port mismatches, and a Cognito token bug
- Stabilized the team's Dockerized development environment by fixing devDependency, platform-binary, and bind-mount conflicts, cutting setup time for a 10-person team from a full day to under 30 minutes

University of Michigan

May 2026 – Present

Undergraduate Research Assistant – ArchSAT: Hardware Performance Analysis for SAT Solvers

Ann Arbor, MI

- Co-developed ArchSAT, a hardware performance framework profiling 3 SAT solvers (Kissat, ParaFROST-CPU/GPU) across 470+ benchmarks with Intel VTune and Linux perf; work targeted for SAT 2027
- Found that absolute memory-operation volume, not cache-efficiency percentage, best predicts solver runtime, surfacing benchmarks where the fastest solver had 6x worse cache miss rates than slower solvers
- Built the end-to-end pipeline from raw profiler traces to an interactive Python/Dash/Plotly dashboard, parsing cache, branch, and BCP metrics and joining them with a SQLite benchmark database for SAT/UNSAT labeling

PROJECTS

Free Food Tracker | React, TypeScript, Express, PostgreSQL | Personal

September 2026 – Present

- Building a web app that aggregates 100+ free-food events per week across the University of Michigan campus into one searchable, time and location based feed so students don't miss them
- Designing a REST API and PostgreSQL schema for events, locations, and sources, with a scraper that ingests postings from 5 campus event feeds

SuperSnakes | Python, PyGame | Personal

May 2025 – Present

- Built an event-driven game with real-time input, collisions, and power-ups at a 10ms input response time, sustaining 60 FPS through controlled frame timing and decoupled rendering in a modular object-oriented design

Chat485: Full-Stack LLM Chat App | Python, Flask, SQLite, React | EECS 485

September 2026 – Present

- Built the Flask backend of a ChatGPT-style app with a 4-table SQLite schema, database-backed UUID session cookies rotated on login, salted SHA-512 password hashing, and 403/404 access control on every conversation
- Integrated an OpenAI-compatible LLM API with a 20-message sliding context window and graceful timeout handling; building a 14-endpoint REST API and a React single-page frontend with optimistic updates and race-safe polling

Pipelined CPU & Cache Simulator | C | EECS 370

Fall 2025

- Built a cycle-accurate 5-stage pipelined processor simulator with data forwarding, load-use stalls, and branch flushes, along with a two-pass assembler and a multi-file linker
- Implemented a configurable cache (block size, sets, associativity) with LRU replacement and write-back, write-allocate policies, logging every block transfer between processor, cache, and memory

TECHNICAL SKILLS

Languages: Python, C++, C, TypeScript, JavaScript, Java, SQL, HTML/CSS, Bash

Frameworks & Libraries: React, Node.js, Express.js, Flask, Tailwind CSS, Dash, Plotly, NumPy, Pandas, scikit-learn

Tools & Platforms: Git, Docker, PostgreSQL, SQLite, AWS Cognito, Stripe, Postman, pytest, Cypress, Linux, VTune